

CHAPTER 1

Tang  
&  
Yang

Rong  
Tang

Yun  
Yang

2019–  
2025

An-  
nals  
of  
Sta-  
tis-  
tics  
JRSS

MCMC

1.

1.1.

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$$D$$

$$\frac{d \ll D}{D}$$

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 $D$ GAN  
VAE

(1)

(2)

(3)

**MCMC**

Tang-  
Yang



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*Tang*  
*&*  
*Yang*  
*2019–*  
*2025*



*MCMC*

*1*

$\mathcal{Q}$   
*VAE*

$\mathcal{Q}$

*MCMC*

**2.**

2.1.

.

$$\mathcal{M}^*$$

$$\mathbb{R}^D$$

$$d$$

$$\beta$$
-

$$\beta \geq 1$$

$$P^*$$

$$\mathcal{M}^*$$

$$\mathrm{vol}_{\mathcal{M}^*}$$

$$p^*$$

$$\alpha-$$

$$\alpha \in [0, \beta-1]$$

$$p^*$$

$$\mathbb{R}^D$$

Lebesgue

————

Lebesgue

Par-  
ti-  
tion  
of  
Unity  
Tang

$$(U_\lambda, \varphi_\lambda)$$

$$\{\gamma_\lambda\}$$

]]tang2022minimax

]]tang2023lowdim

**Reach**

reach



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$$\text{reach}(\mathcal{M}) = \inf\{r > 0 : \mathcal{M}$$

$r$ -

}

**2.2.**

**IPM.**

KL

Hellinger

Tang

IPM

$$d_{\mathcal{F}}(\mu, \nu) = \sup_{f \in \mathcal{F}} \left| \int_{\mathbb{R}^D} f(x) \, \mathrm{d}\mu - \int_{\mathbb{R}^D} f(x) \, \mathrm{d}\nu \right|.$$

$\mathcal{F}$  $\gamma$ -  
Hölder

Sobolev

1-  
LipschitzWasserstein-  
1

RKHS

MMD  
 $\gamma$

trade-  
off

$$\gamma = 1$$

$$d_\gamma$$

Wasserstein-  
1

$$\gamma \rightarrow 0^+$$

$$d_\gamma$$

?]tang2025regression

**2.3.**

•  
Tang

$$= n^{-1/2} \vee n^{-\frac{\alpha+\gamma}{2\alpha+d}} \vee n^{-\frac{\beta\gamma}{d}}.$$

$$n^{-1/2}$$

$$d$$

$\beta$ -

$d$

Haus-  
dorff

$D$





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Gir-  
sanov

Poincaré

2.4.

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Gauss-  
ian

Langevin

$$p_{\sigma}(x) = \int p_{\text{data}}(y) \phi_{\sigma}(x - y) \, \mathrm{d}y$$

$$L_{\infty}$$

-

$$\nabla \log p_t(x)$$

$$t$$

—

$$t =$$

$$0$$

$$\frac{t}{T} =$$

?]tang2024diffusion

$$\mathcal{S}_{\mathrm{NN}}$$

$$t$$

**Girsanov**

Gir-  
sanov

SDE

3.

3.1. ?]]tang2022minimax

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Rong  
Tang  
and  
Yun  
Yang.

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2022.



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 $D$ 

GAN  
VAE



$\gamma$ 

$$n^{-1/2} \vee n^{-\frac{\alpha+\gamma}{2\alpha+d}} \vee n^{-\frac{\beta\gamma}{d}},$$

 $\alpha$  $\beta$  $d$

$D$

par-  
ti-  
tion  
of  
unity

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**VAE**

Taylor

Lepski

Fano

Varshamov-Gilbert

]]tang2022minimax

]]tang2023lowdim,  
]]tang2024diffusion

### 3.2. ?]]tang2023lowdim

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Rong  
Tang  
and  
Yun  
Yang.



“Estimating Distributions with Low-dimensional Structures Using Mixtures of Generative Models.” *ICML*, 2023.

VAE  
GAN

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MLD-  
MAE

$K$

-

$\rho_k$

**1-**  
**Wasserstein**

$\alpha$ -

$\beta$ -

$d$

$$O\Big(n^{-\frac{\alpha\wedge(\beta-1)+1}{2(\alpha\wedge(\beta-1))+d}}\Big)\vee O\Big(\frac{\log n}{\sqrt{n}}\Big),$$

$$\begin{array}{l} \alpha \leq \\ \beta - \\ 1 \end{array}$$

$$/$$

?]tang2022minimax

?]tang2022minimax

### 3.3. ?]]tang2023robust

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Rong  
Tang,  
Anir-  
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*JRSS*  
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2023.



Stiefel

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Bayesian  
RPE-  
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(1)

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Bernstein-  
von  
Mises  
(BvM)



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3.4. ?]]tang2023twosample

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Rong  
Tang  
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2023.

**MMD**

$O(n^2)$

Besov

Besov

**1** $\ell_1$   
 $\ell_2$ **Wasser-  
stein**

1

2

$$O(n \log n + n^{2d/(4\alpha+d)})$$

MMD

$$O(n^2)$$

3.5. ?]]tang2024diffusion

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TATS*,  
2024.

**SOTA**

**Langevin**

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$d$

$D$   
**Langevin**



Gauss-  
ian

$\tilde{O}(\sigma)$

$\sigma$

$n$

$d$

$L_\infty$ 

—

1-  
Wasserstein

$$\frac{1}{\sqrt{n}} \vee n^{-\frac{\alpha+1}{2\alpha+d}}.$$

?]tang2022minimax

?]tang2022minimax

### 3.6. ?]]tang2024mala

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Metropolis-  
Adjusted  
Langevin

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Rong  
Tang  
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*ICML*,  
2024.

MALA

MCMC

$$O(d^{1/3})$$

*s*-  
conductance  
pro-  
file

**MALA**

$O(d^{1/3})$

**burn-  
in**



s-  
conductance  
pro-  
file

con-  
duc-  
tance

Chen  
et  
al.  
(2020)

con-  
duc-  
tance  
pro-  
file

warm-  
start

$\log M_0$

$\log \log M_0$

?]tang2023robust

?]tang2023robust

BvM

BvM

MALA

3.7. ?]]tang2025regression

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arXiv,  
2025.

$X$

$Y$

$\mu_{Y|X}^*$

$X$

$Y$



IPM

(1)  
Regime  
1

**Lebesgue**

$$n^{-\frac{\alpha_X}{2\alpha_X+d_X}}+n^{-\frac{\alpha_Y+\gamma}{2\alpha_Y+D_Y+\frac{\alpha_Y}{\alpha_X}\cdot d_X}}.$$

**(2)**

**Regime**  
**2**

:

$\mathcal{M}_{Y|X}$

$X$



(3)  $n^{-\gamma\beta_Y/d_Y}$   
**Regime**  
**3**

:

$\mathcal{M}_{Y|x}$

$x$

$n^{-\gamma/(\frac{d_Y}{\beta_Y}+\frac{d_X}{\beta_X})}$

$\gamma$ -  
**Hölder**

**IPM**

$d_\gamma$

$\gamma =$   
0

**1-**  
**Wasserstein**

$\gamma =$   
1

**3.8. ?]]tang2025conditional**  
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**Rong**  
**Tang**  
**and**  
**Yun**  
**Yang.**

“Con-  
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Mod-  
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are  
Minimax-  
Optimal  
and  
Manifold-  
Adaptive,”  
arXiv,  
2025.

?]]tang2024diffusion



(1)

$$\alpha_X \in [0, 1]$$

(2)

$X$ 
 $Y$ 

**Wasser-  
stein**

 $(d_X, d_Y)$ 
 $(D_X, D_Y)$

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niques

Gir-  
sanov



?]tang2024diffusion,  
?]tang2025regression

?]tang2024diffusion

→  
?]tang2025regression

→

4.

Tang-  
Yang

2019—  
2025

4.1.